

Remote Desktop Protocol (RDP) Setup & Network Level Authentication (NLA)

Secure RDP Deployment: Configure Windows/Linux RDP services and enforce Network Level Authentication (NLA) for pre-session validation.

Introduction

Remote Desktop Protocol (RDP) is a remote access technology that allows users to connect to and control another computer through a graphical desktop interface over a network. It is widely used for remote system administration, technical support, and accessing resources from different locations. To improve the security of remote connections, Network Level Authentication (NLA) verifies user credentials before a full remote desktop session is established. This reduces the risk of unauthorized access, brute-force attacks, and denial-of-service attempts while ensuring that only authenticated users can access the remote system.

Objectives

Aim: To configure and secure a Remote Desktop Protocol (RDP) service on a Windows/Linux system by enabling Network Level Authentication (NLA), and to establish a secure remote desktop connection while verifying successful authentication through system audit logs. Enable RDP access and configure allowed user group policies.

Requirements

- **Target Machine:** Windows 10/11, Windows Server, or Linux system with **xrdp** installed and configured.
- **Client Machine:** Windows or Linux system with an RDP client such as **Remote Desktop Connection (MSTSC)** or **Remmina**.
- **Network Connectivity:** Both machines are connected to the same network or reachable over IP.
- **Administrator Privileges:** Required to enable RDP and configure Network Level Authentication (NLA).
- **Valid User Credentials:** Authorized username and password for remote login.
- **Event Logging Enabled:** Windows Event Viewer or Linux authentication logs (/var/log/auth.log) available for verification.

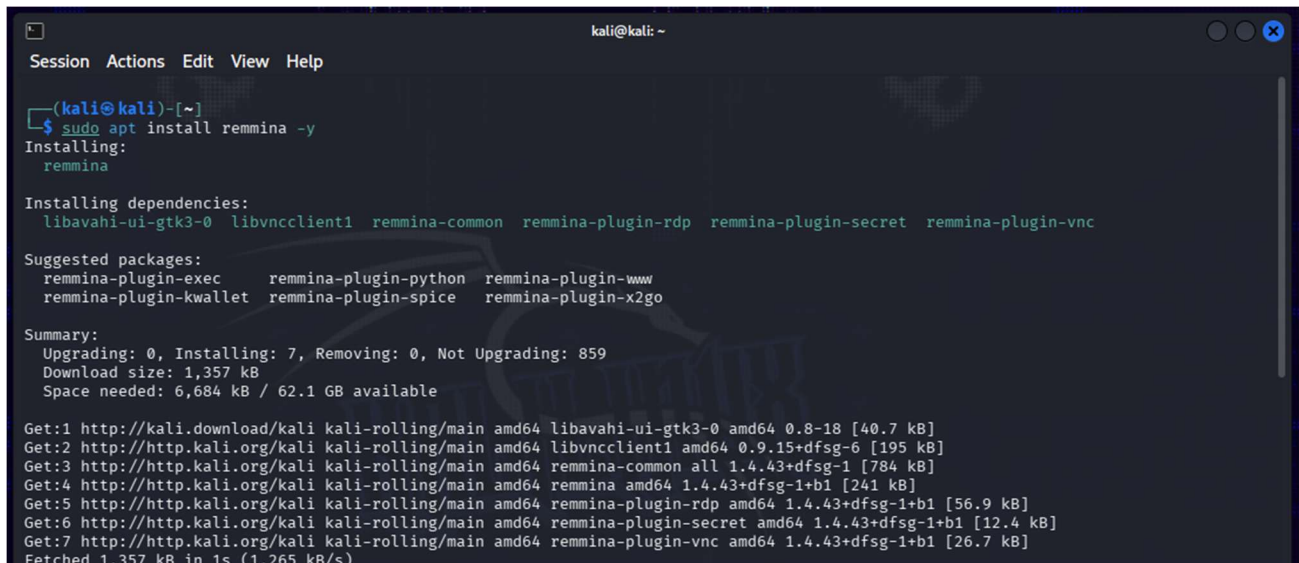
Expected Output

- Remote Desktop Protocol (RDP) service is successfully enabled on the target machine.
- Network Level Authentication (NLA) is enforced for secure pre-session user authentication.
- A secure graphical remote desktop session is established from the client machine using valid credentials.
- Successful authentication and session initiation are recorded in the Windows Event Viewer (Event ID 4624) or Linux authentication logs.
- The remote desktop can be accessed securely while preventing unauthorized connections without valid authentication.

Step by Step Procedure(s)

Step-1: Install the Remmina RDP Client on Kali Linux

Open the terminal on the Kali Linux machine and update the package repository. Install the **Remmina** Remote Desktop client using the APT package manager. Wait until the installation completes successfully without any errors before proceeding to the next step.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ sudo apt install remmina -y  
Installing:  
  remmina  
  
Installing dependencies:  
  libavahi-ui-gtk3-0 libvncclient1 remmina-common remmina-plugin-rdp remmina-plugin-secret remmina-plugin-vnc  
  
Suggested packages:  
  remmina-plugin-exec remmina-plugin-python remmina-plugin-ssh remmina-plugin-xfreerdp  
  remmina-plugin-kwallet remmina-plugin-spice remmina-plugin-x2go  
  
Summary:  
  Upgrading: 0, Installing: 7, Removing: 0, Not Upgrading: 859  
  Download size: 1,357 kB  
  Space needed: 6,684 kB / 62.1 GB available  
  
Get:1 http://kali.download/kali kali-rolling/main amd64 libavahi-ui-gtk3-0 amd64 0.8-18 [40.7 kB]  
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libvncclient1 amd64 0.9.15+dfsg-6 [195 kB]  
Get:3 http://http.kali.org/kali kali-rolling/main amd64 remmina-common all 1.4.43+dfsg-1 [784 kB]  
Get:4 http://http.kali.org/kali kali-rolling/main amd64 remmina amd64 1.4.43+dfsg-1+b1 [241 kB]  
Get:5 http://http.kali.org/kali kali-rolling/main amd64 remmina-plugin-rdp amd64 1.4.43+dfsg-1+b1 [56.9 kB]  
Get:6 http://http.kali.org/kali kali-rolling/main amd64 remmina-plugin-secret amd64 1.4.43+dfsg-1+b1 [12.4 kB]  
Get:7 http://http.kali.org/kali kali-rolling/main amd64 remmina-plugin-vnc amd64 1.4.43+dfsg-1+b1 [26.7 kB]  
Fetched 1,357 kB in 1s (1,265 kB/s)
```

Fig 3.1: installation of remmina

Step-2: Enable Remote Desktop and Network Level Authentication on Windows

On the Windows target machine, navigate to Settings → System → Remote Desktop and enable the Remote Desktop option. Ensure that Require devices to use Network Level Authentication (NLA) is checked to enforce secure pre-authentication before allowing remote desktop connections.

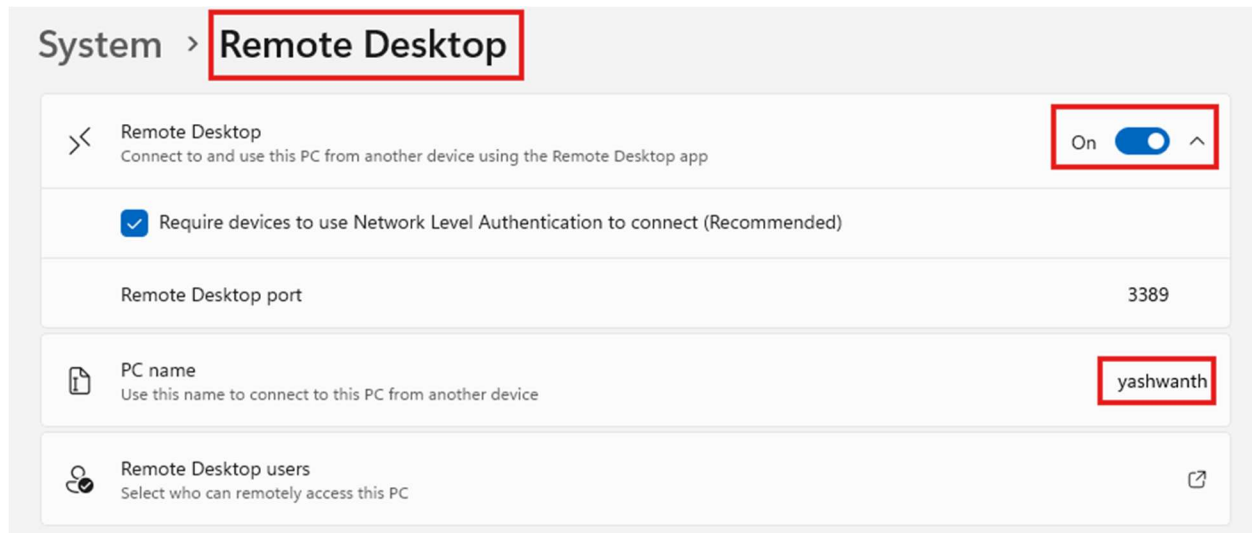


Fig 3.2 Enabling RDP on windows system

Step-3: Launch the Remmina Remote Desktop Client

After installing Remmina, open the application from the Kali Linux applications menu by searching for **Remmina**. Verify that the application launches successfully and is ready to create a new Remote Desktop Protocol (RDP) connection.

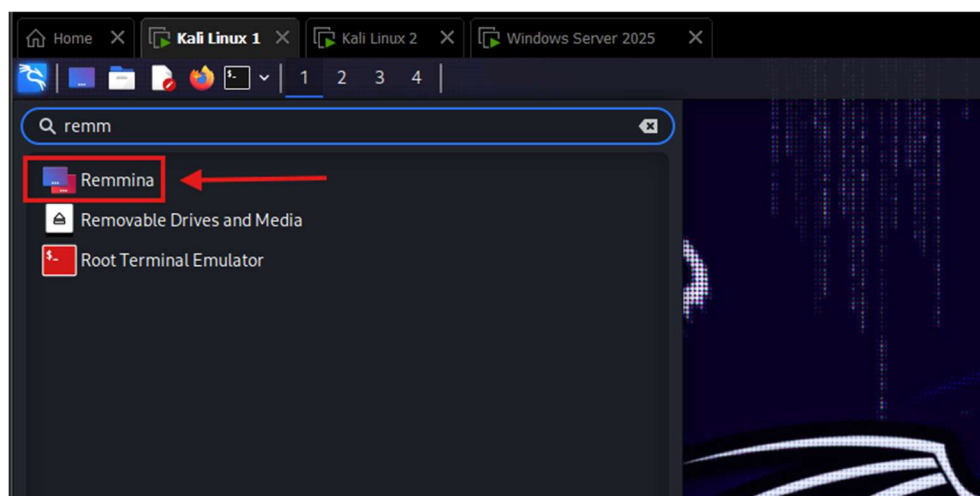


Fig 3.3: Launch Remmina

Step-4: Configure the RDP Connection and Authenticate

In Remmina, select RDP as the connection protocol and enter the Windows machine's IP address. Provide the valid Windows username and password, then click OK to authenticate and initiate the secure remote desktop connection.

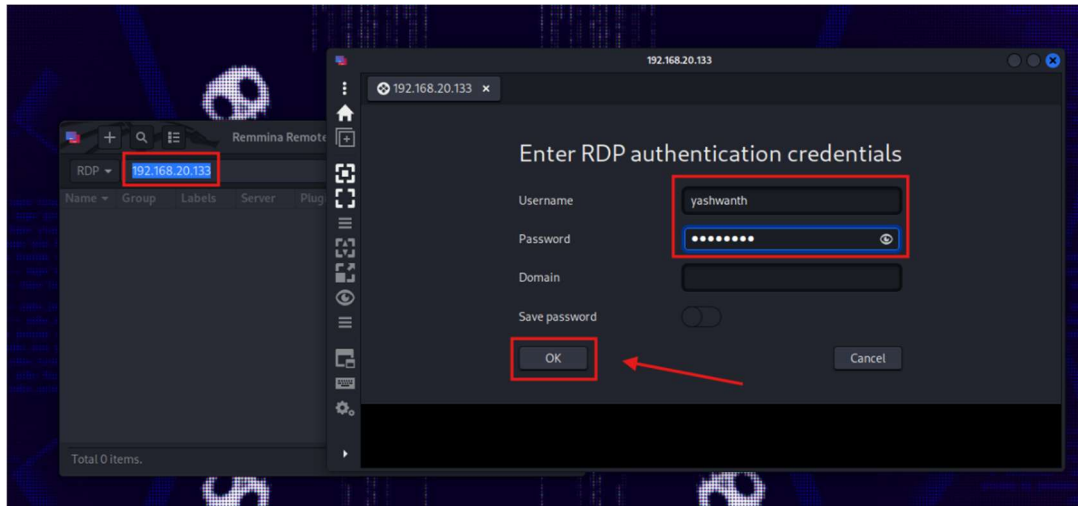


Fig 3.4: enter ip address and give credentials

Step-5: Verify Successful Remote Desktop Connection

After successful authentication, confirm that the Windows desktop is displayed inside the Remmina window on the Kali Linux machine. Verify that the connection has been established successfully and that the remote Windows system can now be accessed and managed through the secure RDP session.

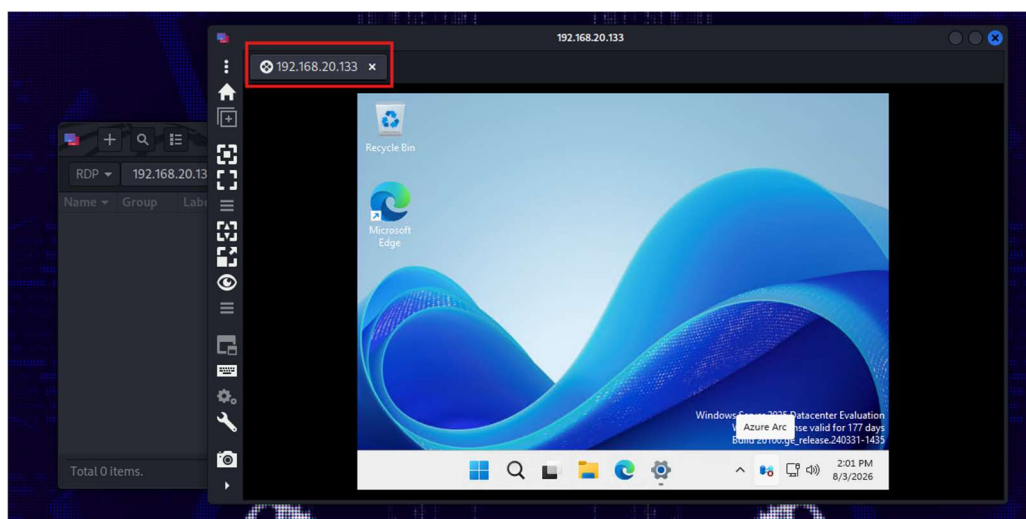


Fig 3.5: RDP Connection Success

Learning Outcomes

- Understand the configuration and secure deployment of Remote Desktop Protocol (RDP) services.
- Learn how Network Level Authentication (NLA) protects remote desktop connections through pre-session user authentication.
- Gain hands-on experience in establishing a secure remote desktop connection from a Kali Linux client to a Windows host using Remmina.
- Verify successful remote authentication and session establishment using Windows security logs.
- Develop practical skills in securely managing and accessing remote systems over a network. DP Connection

Notes/Observations

- The Remmina Remote Desktop client was successfully installed on the Kali Linux machine and used to establish an RDP connection.
- Remote Desktop was enabled on the Windows target system with **Network Level Authentication (NLA)** enforced for secure access.
- The client authenticated successfully using valid Windows credentials before the graphical desktop session was established.
- The remote Windows desktop was displayed correctly on the Kali Linux machine, confirming successful RDP communication.
- The connection remained stable throughout the session, allowing normal interaction with the remote system.
- Authentication and login activities were recorded in the Windows Security Event Logs, confirming successful session establishment.
- Enabling NLA provided an additional layer of security by preventing unauthenticated users from initiating a full remote desktop session.
- This experiment demonstrated the practical implementation of secure remote desktop access between Windows and Kali Linux in a networked environment.